# VLSI Devices
# Lecture 7

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# L7

# MOS capacitor

- Voltage across the capacitor, $V_g$
  - Silicon charge, $Q_s$
  - Gate charge, $Q_g = -Q_s$
  - Therefore, we can consider

$$-Q_s(V_g)$$

# An ideal capacitor

- Consider a planar capacitor.
  - Its dielectric layer is $SiO_2$. It is 2-nm-thick.
  - Then,

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = 1.73\ \mu\mathrm{F/cm}^2$$

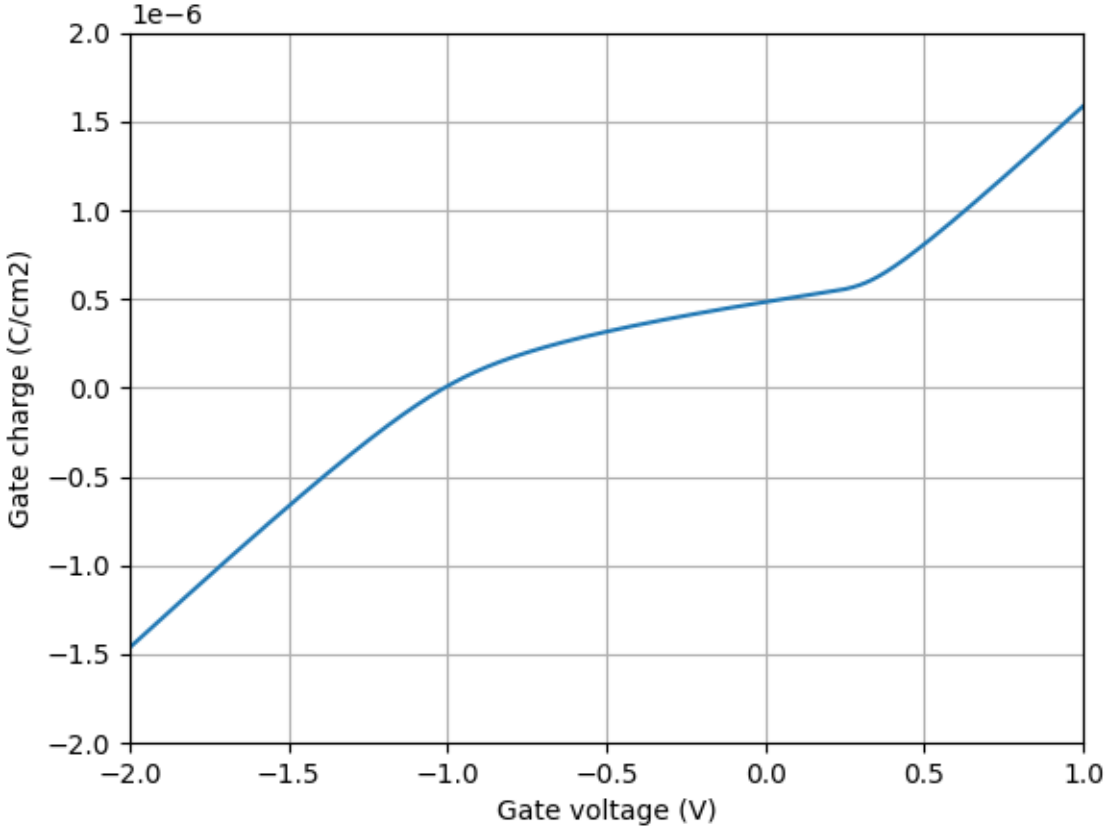
  - Draw its $Q(V)$ curve.

# $-Q_s(V_g)$ of a MOS capacitor

- We cannot find a capacitance value for all biases. Instead,

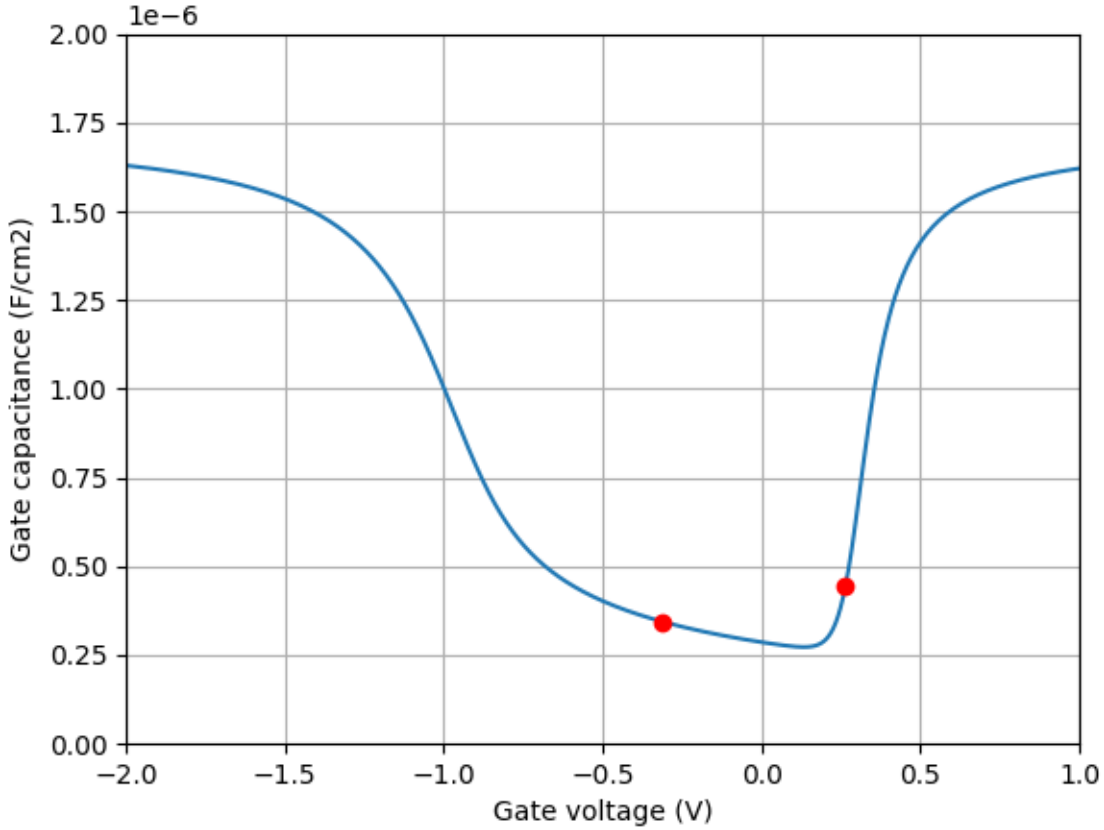
$$C_g = \frac{dQ_g}{dV_g} = -\frac{dQ_s}{dV_g}$$

Taur, Eq. (4.34)



# Just draw $C_g$.

- Bias-dependent capacitance
  - Remember that $C_{ox} = 1.73\ \mu\mathrm{F/cm^2}$.
  - Frequency?



# Expressions

- Remember the MOS equation.

$$V_g - V_{fb} = \frac{Q_g}{C_{ox}} + \phi_s$$

Taur, ~Eq. (4.7)

  - Differentiation w.r.t. $Q_g$ yields

$$\frac{dV_g}{dQ_g} = \frac{1}{C_{ox}} + \frac{d\phi_s}{dQ_g}$$

  - We obtain

$$\frac{1}{C_g} = \frac{dV_g}{dQ_g} = \frac{1}{C_{ox}} + \frac{d\phi_s}{dQ_g} = \frac{1}{C_{ox}} + \frac{1}{C_{si}}$$

Taur, Eq. (4.36)

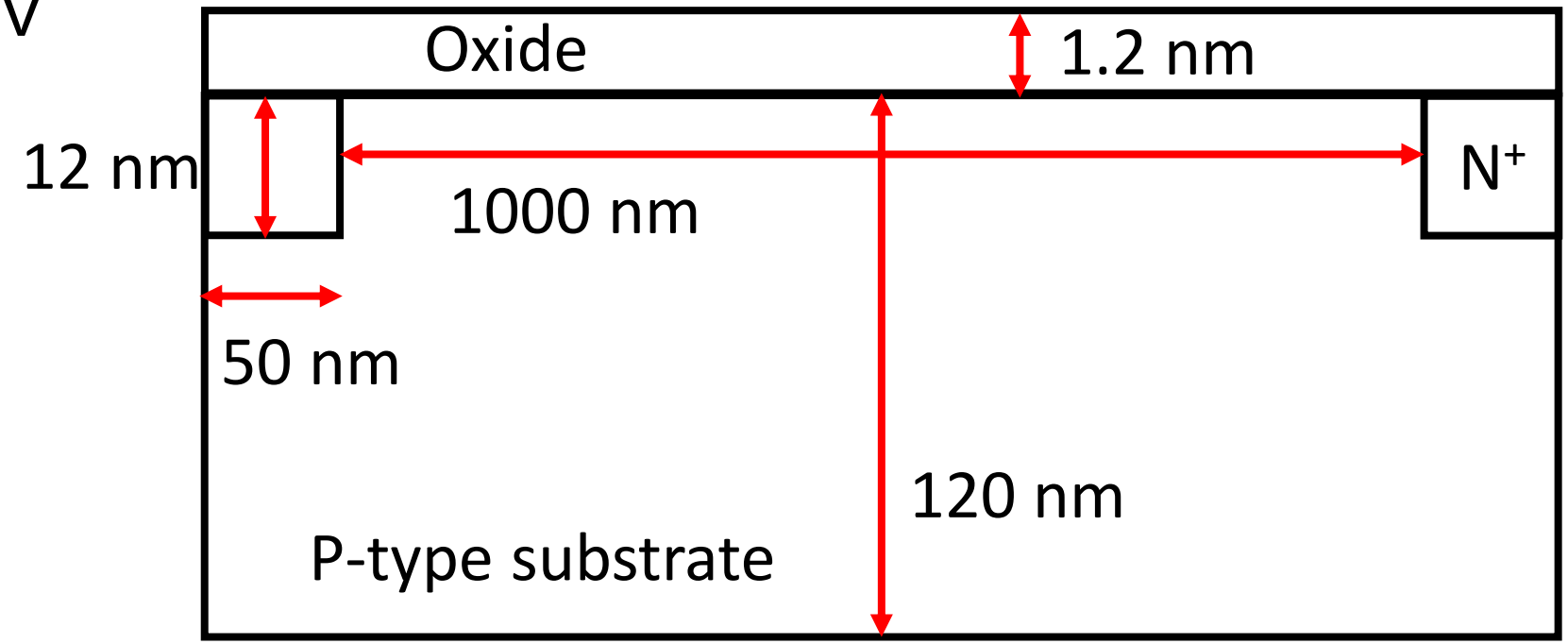
$$C_{si} = \frac{dQ_g}{d\phi_s} = -\frac{dQ_s}{d\phi_s}$$

# Homework#7

- Due: 08:00 on March 30
- Submit a PDF report through the GIST LMS system.
  - We know $Q_s(\phi_s)$. Derive an explicit expression for $C_{si}$.

# Example of a long-channel MOSFET

- Effective oxide thickness of 1.2 nm
  - Gate workfunction of 4.3 eV
  - Substrate doping of $1.5X10^{18}$ $cm^{-3}$
  - $V_{dd}$ of 1.2 V

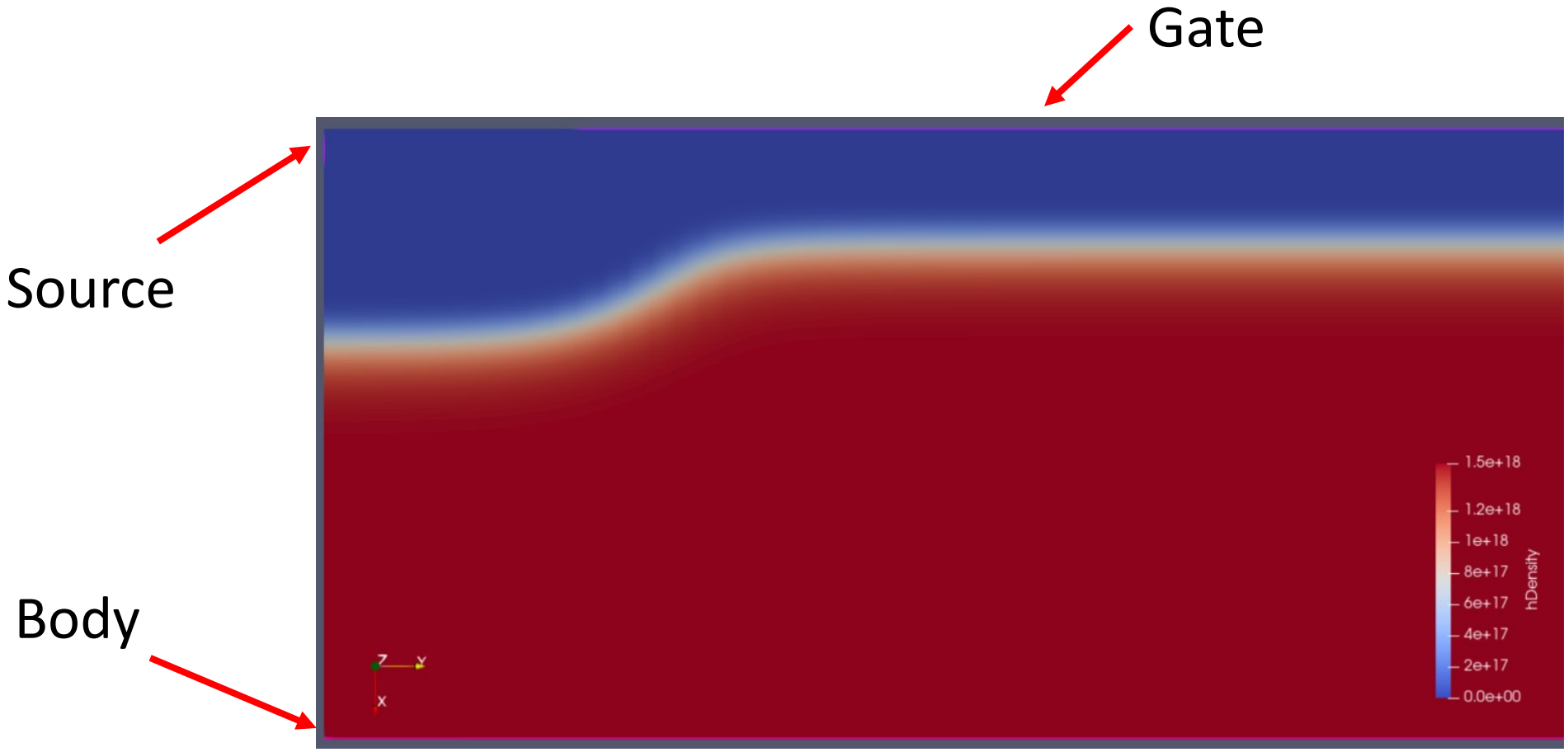


# Computer program

- For accurate prediction of electrical characteristics
  - Gate workfunction of 4.3 eV

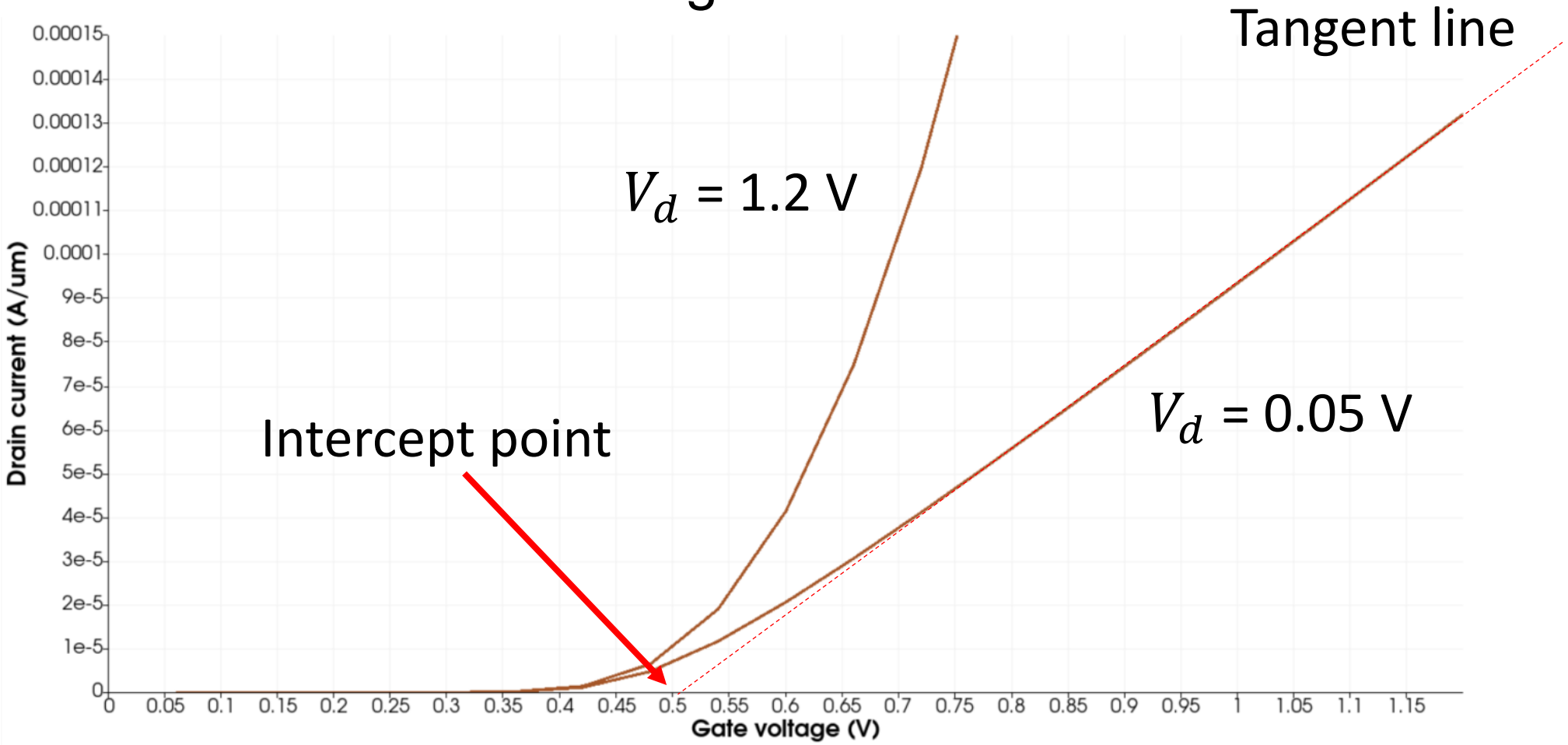
# Hole density at $V_g = V_d = 0$ V

- Depletion of holes



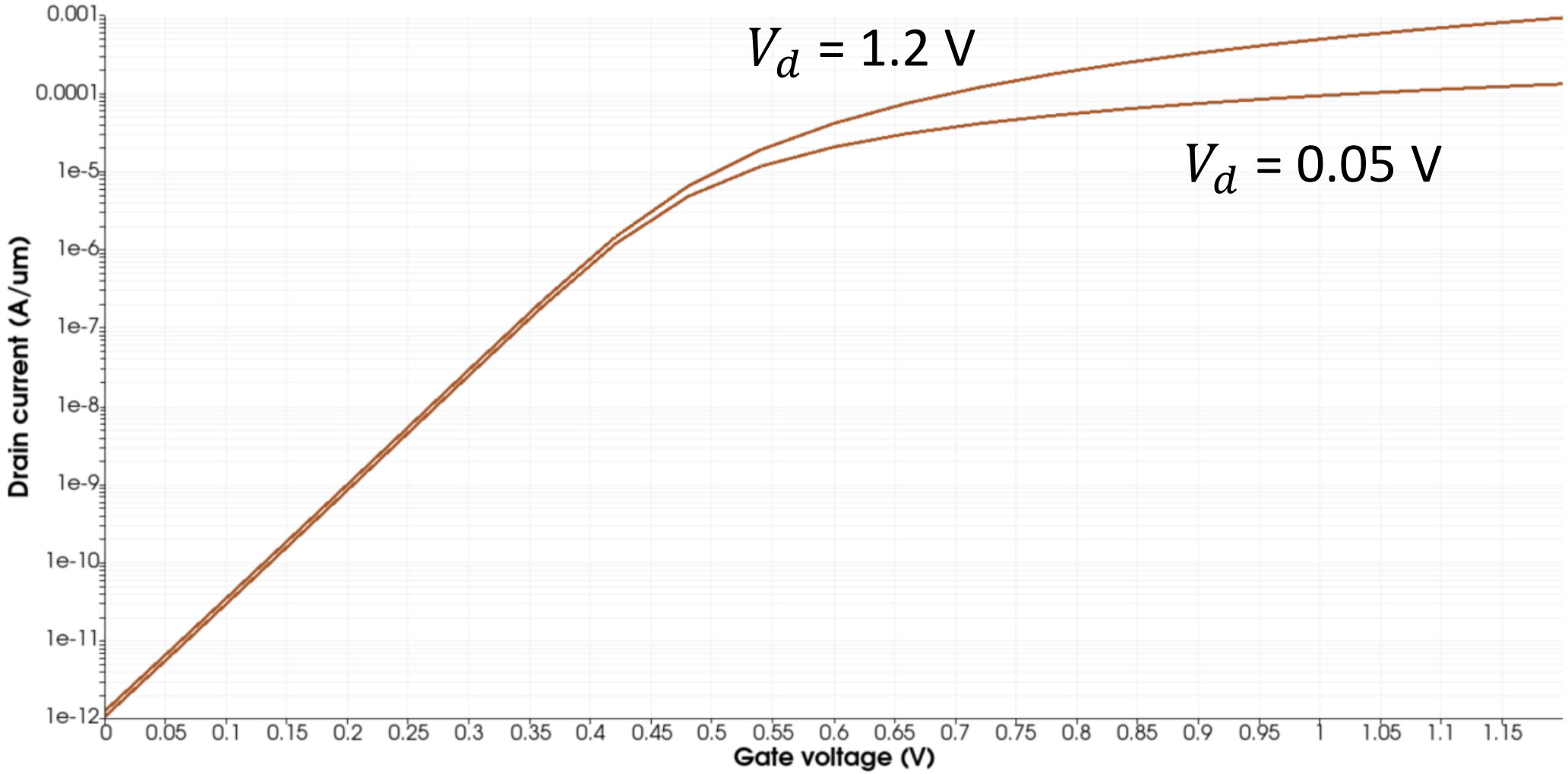
# Ramping the gate voltage up

- From 0 V to 1.2 V ($V_{dd}$)
  - Estimate the threshold voltage.



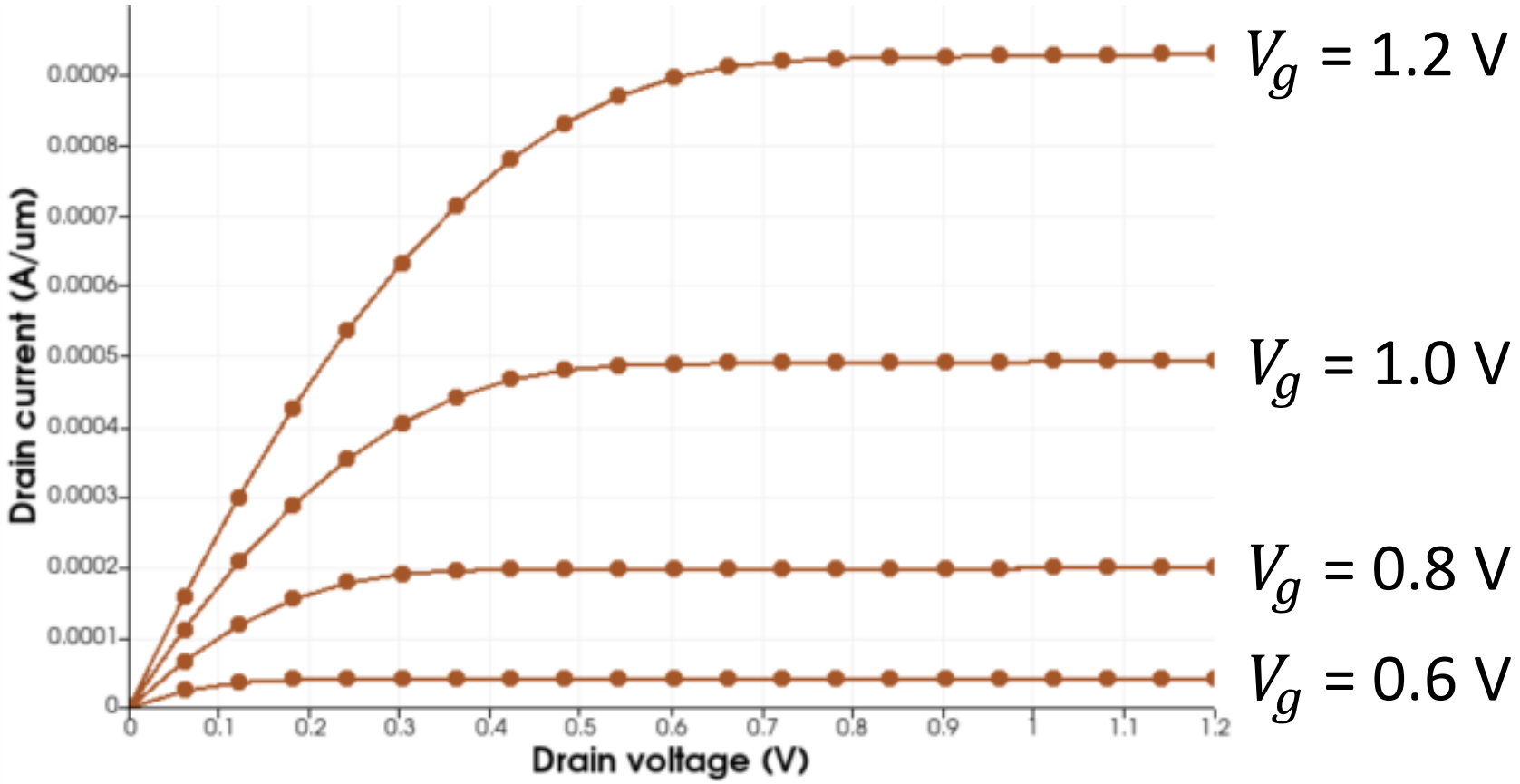
# Semi-logarithmic graph

- Estimate the subthreshold slope, the DIBL, $I_{ON}$, and $I_{OFF}$.



# Output characteristics

- Results from $V_g$ = 0.6 V to 1.2 V (0.2 V step)



# Thank you!